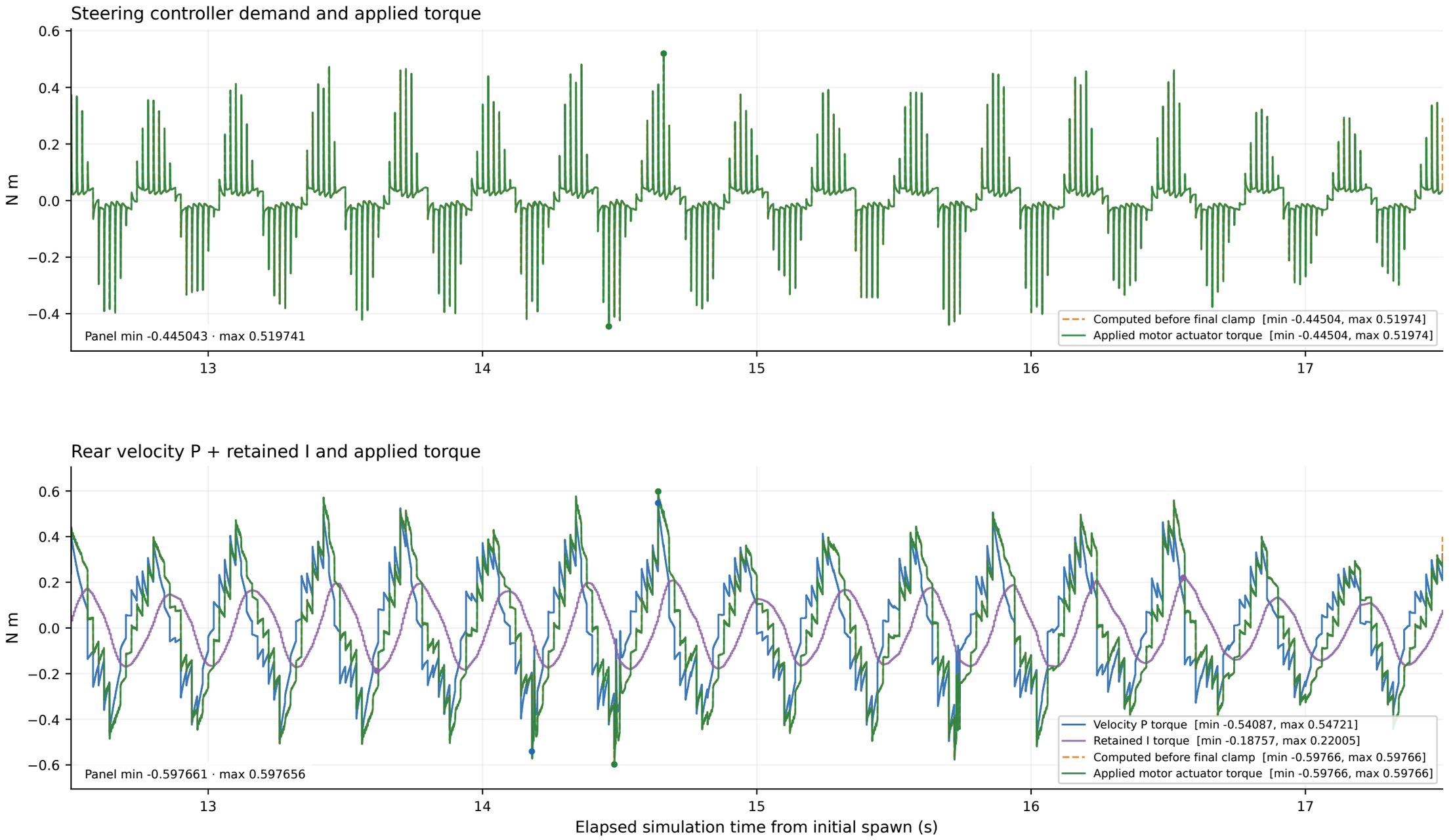



Nominal fixed reset; ordinary IMU white noise retained. Red: initial spawn at 0 s. No second episode has recorded samples.
Steering position/speed: 50 Hz boundaries; rear feedback/torques: 1200 Hz. Applied motor torque excludes passive friction/contact joint effort.
Fixed gains: steering 10 / 0.2; rear velocity P 0.30, Ki 3.3. Nominal tyre friction 0.65; rear frictionloss 0.106 N m, passive B 0.